INVESTMENTS

Ninth Canadian Edition · Bodie, Kane, Marcus

CHAPTER 12

Behavioural Finance and Technical Analysis

Verified against source text — learning objectives, formulas, and worked examples checked directly against the chapter.

Study Guide · Prepared for Personal Use

LO1 · The Behavioural Critique

Behavioural finance argues that conventional financial theory (including the EMH from Chapter 11) ignores **how real people actually make decisions.** The behavioural critique rests on two related claims: (1) real investors are subject to systematic **information-processing errors**, producing biased estimates of probabilities or expected returns, and (2) even when investors process information correctly, they may make systematically different **decisions** given identical information and beliefs — that is, irrationality can enter through faulty inputs, faulty processing of correct inputs, or both.

Information Processing Biases

Bias	Description
Forecasting Errors	People tend to overweight recent experience relative to prior (base-rate) beliefs and produce forecasts that are too extreme given the actual uncertainty involved.
Overconfidence	People overestimate the precision of their own beliefs and abilities — e.g., most drivers rate themselves better than average. In markets, overconfident investors trade more (evidence shows men, on average, trade more actively than women, consistent with greater documented overconfidence) and may overpay for acquisition targets.
Conservatism	Investors are too slow to update beliefs in response to new evidence, underreacting to genuine changes in fundamentals.
Representativeness (Sample-Size Neglect)	People judge probabilities by how much a situation "resembles" a familiar category, without properly weighting the number of observations — e.g., inferring a firm has a durable growth trend from just a few years of strong earnings.

Behavioural Biases (Errors in How Decisions Are Made)

Even given correctly processed information, actual decisions can deviate from those predicted by conventional rational choice theory:

Bias	Description
Framing	The same objective choice can produce different decisions depending on how it is presented (framed) — e.g., describing an outcome in terms of gains versus losses can flip an investor’s willingness to accept a given risky bet.
Mental Accounting	A specific form of framing: people segregate decisions into separate mental "accounts" and analyze them individually rather than considering their combined effect on total wealth — e.g., feeling comfortable taking a risky bet with "house money" (recent gains) that they would not take with their original capital.
Regret Avoidance	Investors feel more regret for a bad outcome that resulted from an unconventional or unusual decision than for an equally bad outcome from a conventional decision — which can help explain the market’s tendency to prefer conventional, well-known ("blue chip") stocks and the P/E effect discussed in Chapter 11.
The Disposition Effect	The well-documented tendency of investors to hold on to losing investments too long (reluctant to realize a loss) while selling winners too early. This effect can itself generate price momentum, even if a security’s true fundamental value follows a pure random walk.

> *The dot-com boom is often cited as a case study combining several biases at once: investors grew increasingly confident in their own investing prowess (overconfidence) while extrapolating short recent price trends far into the future (representativeness), a self-reinforcing pattern until it reversed.*

Limits to Arbitrage

If behavioural biases cause some investors to misprice securities, why don't rational arbitrageurs simply trade against them and correct the mispricing (as the EMH would predict)? Behavioural finance argues several real-world frictions limit this correction mechanism:

- **Fundamental risk —** a security that looks mispriced might become even more mispriced before it corrects (or might reflect a genuine, currently unknown risk), so a bet against it is never truly riskless.
- **Implementation costs —** short-selling costs, restrictions on who can short, and difficulty borrowing shares all raise the cost of correcting overpricing specifically.
- **Model risk —** an arbitrageur can never be fully certain that a security is actually mispriced rather than that their own valuation model is the one that is wrong.

Limits to Arbitrage and the Law of One Price — "Siamese Twin" Companies

The clearest evidence that arbitrage limits are real, not merely theoretical: several cases exist where the **Law of One Price** (identical assets should have identical prices) was directly and persistently violated. In 1907, Royal Dutch Petroleum and Shell Transport merged operations but continued trading as two separate shares, contractually splitting all profits 60/40. Royal Dutch should therefore always trade at exactly 60/40 = 1.5 times Shell's price — but it did not: in February 1993, Royal Dutch traded about **10% above** its fair parity value, and that premium **widened to about 17%** before finally reversing after 1999. Any investor who tried to arbitrage the initial 10% mispricing (buying underpriced Shell, shorting overpriced Royal Dutch) would have suffered significant losses as the gap widened further — a textbook illustration of fundamental risk limiting real-world arbitrage.

A related case: 3Com's 1999 spinoff of its Palm division. Prior to the full spinoff (in which each 3Com shareholder would receive 1.5 Palm shares), 3Com's own share price should mechanically have been at least 1.5 times Palm's — yet for a period after Palm began trading, this relationship was violated, implying the market was implicitly assigning a **negative value** to the rest of 3Com's substantial ongoing business.

LO2 · Technical Analysis and Behavioural Finance

Technical analysis attempts to exploit recurring, predictable patterns in prices and trading volume. Technicians do not deny that fundamental (intrinsic) value matters — they believe prices only **gradually** converge to that intrinsic value, creating exploitable patterns along the way. Notably, several behavioural biases from LO1 provide a possible rationale for *why* such patterns might genuinely exist: the disposition effect (reluctance to sell losers) can generate real price momentum even when fundamentals follow a pure random walk, and overconfidence-driven trading can link volume data to subsequent returns — which is exactly why technicians track volume alongside price.

Trends, Momentum, and Moving Averages

Much of technical analysis searches for **momentum** — either absolute (an upward price trend) or relative (one sector or stock outperforming another). A **moving average** (e.g., a 50-day average, continuously updated as time passes) smooths out short-term noise to reveal an underlying trend. After a period of falling prices, the moving average sits *above* the current price (since it still reflects some now-stale higher prices); after a sustained rise, it sits *below* the current price. A price breaking through its moving average from below is read as a shift from a falling to a rising trend, and vice versa.

Relative Strength

Relative strength statistics compare a security's (or sector's) performance directly against a broader benchmark (e.g., Toyota's return versus the overall auto industry's return) rather than looking at absolute price levels. If relative strength persists over time, technicians look to invest in the stronger-performing sector, sometimes via a long-short position across the two.

Breadth and the Trin Statistic

Market **breadth** measures how broadly a market move is shared across individual stocks, typically as the spread between the number of advancing versus declining stocks on a given day, often cumulated over time to reveal underlying trend strength. The **trin statistic** (a ratio comparing volume in advancing versus declining stocks) is a related breadth measure; a trin value above 1.0 is conventionally read as bearish (more volume is flowing into declining stocks relative to advancing ones).

A Critical Perspective on Technical Analysis

Skeptics note that many technical signals are defined loosely enough (with hindsight-friendly labels evoking drama — "death cross," "Hindenburg omen," "negative key reversal") that they can be fitted to almost any chart after the fact, raising concerns about **data mining:** testing enough patterns against historical data will eventually turn up some that appear statistically significant purely by chance, with no genuine predictive power going forward.

At the same time, technical analysis and behavioural finance are increasingly viewed as complementary rather than opposed: behavioural finance offers a plausible *mechanism* (systematic investor biases like the disposition effect and overconfidence) for why the price patterns technicians have long claimed to observe could genuinely persist, rather than dismissing technical analysis outright as pure superstition.

Chapter 12 — Big-Picture Takeaways

- Behavioural finance challenges the EMH on two fronts: investors may process information with systematic biases, and even correctly processed information can lead to systematically biased decisions.
- Information-processing biases (overconfidence, conservatism, representativeness, forecasting errors) and decision biases (framing, mental accounting, regret avoidance, the disposition effect) together offer behavioural explanations for many of Chapter 11’s anomalies.
- Limits to arbitrage (fundamental risk, implementation costs, model risk) explain why rational traders may not fully correct behaviourally driven mispricing — dramatically illustrated by real, sustained Law-of-One-Price violations (Royal Dutch/Shell, 3Com/Palm).
- Technical analysis searches for exploitable price and volume patterns (moving averages, relative strength, breadth, trin); behavioural biases like the disposition effect offer a possible explanation for why such patterns might genuinely exist.
- A key critique of technical analysis is data mining risk: with enough patterns tested against history, some will appear to "work" by pure chance.
- Behavioural finance and technical analysis increasingly inform one another: behavioural biases provide a candidate mechanism for the very patterns technicians have long claimed to observe.

This chapter is conceptually oriented, but two quantitative relationships are commonly tested.

F1 | Law-of-One-Price Parity Ratio ("Siamese Twin" Shares)

$$\text{Fair Price Ratio} = \text{Split A} / \text{Split B}$$

$$\text{Actual Premium (or Discount)} = (\text{Actual Ratio} - \text{Fair Ratio}) / \text{Fair Ratio}$$

For a profit-sharing agreement splitting cash flows A%/B% between two separately traded shares, the fair price ratio should equal A/B exactly (e.g., 60/40 = 1.5 for Royal Dutch vs. Shell). Persistent deviations from this ratio are direct evidence of a Law-of-One-Price violation.

F2 | Moving Average (Simple, n-Day)

$$MA_t = (P_t + P_{t-1} + \dots + P_{t-n+1}) / n$$

The average closing price over the trailing n days, recalculated (rolled forward) each new trading day. Price crossing above/below the moving average is read by technicians as a potential trend-change signal.

F3 | Trin Statistic (Volume Breadth)

$$\text{Trin} = \frac{\text{Volume on Declining Stocks} / \text{Number of Declining Stocks}}{\text{Volume on Advancing Stocks} / \text{Number of Advancing Stocks}}$$

Trin > 1.0 is conventionally read as bearish (heavier volume concentrated in declining issues relative to advancing ones); Trin < 1.0 is read as bullish.

EXAMPLE | Royal Dutch / Shell — A Law-of-One-Price Violation

Given: Royal Dutch and Shell agreed to split all profits from their merged operations 60%/40%.

Step 1 — Fair price ratio:

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Fair Ratio = 60 / 40 = 1.5   (Royal Dutch should trade at 1.5 × Shell’s price)
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Step 2 — Actual observed deviation (February 1993 onward):

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Initial premium ≈ 10% above fair value   →   widened to ≈ 17%   →   reversed only after 1999
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An arbitrageur shorting overpriced Royal Dutch and buying underpriced Shell in February 1993 (expecting convergence) would have suffered a widening loss for over six years before the position finally became profitable — a textbook demonstration of fundamental risk as a genuine limit to arbitrage, not just a theoretical possibility.

EXAMPLE | Reading a Moving-Average Signal

Given: A stock has been falling for several weeks; its 50-day moving average currently sits above its current price. This week, the price rises sharply and crosses above the 50-day moving average.

Interpretation: a technician would read this crossover as a signal that the stock is shifting from a falling trend to a rising trend (point "A" type breakout in the textbook’s terminology) — a classic buy signal in technical analysis. A behavioural finance proponent might add: if this reversal partly reflects investors finally capitulating and selling long-held losing positions (overcoming the disposition effect), the moving-average crossover could be capturing a genuine, behaviourally driven shift in supply and demand rather than a purely arbitrary pattern.

PRACTICE PROBLEM | Sarcee Ridge Capital

Scenario

You are an analyst at Sarcee Ridge Capital in Calgary. Two Canadian energy companies, Bragg Creek Resources and Millarville Energy, merged their operations years ago under a permanent 70%/30% profit-sharing agreement, but continue to trade as two separate TSX-listed shares.

Part A — Behavioural Biases in Client Behaviour

A client refuses to sell a losing position in Bragg Creek Resources, saying "I'll just wait until it gets back to what I paid."

(i) Name the specific behavioural bias this illustrates.

(ii) A second client made an unusual, contrarian investment last year that lost money, and now says they regret the decision far more than an equally-sized loss from a "safe," conventional stock. Name this bias.

(iii) Briefly explain how bias (ii) could help explain why many investors cluster into well-known "blue chip" stocks.

Part B — The Law of One Price

(i) Given the 70%/30% profit split, calculate the fair price ratio between Bragg Creek Resources and Millarville Energy.

(ii) Suppose Bragg Creek is currently trading at a price that implies a ratio 12% above this fair value. Is Bragg Creek overpriced or underpriced relative to Millarville?

(iii) Explain why a portfolio manager might hesitate to arbitrage this gap immediately, referencing the Royal Dutch/Shell case.

Part C — Technical Analysis Tools

Millarville Energy's 50-day moving average has been above its current price for the past month, but this week the price crossed above the moving average.

(i) What trend-change signal would a technical analyst read into this crossover?

(ii) On the crossover day, advancing stocks on the TSX had much heavier volume than declining stocks. Would the resulting trin statistic be above or below 1.0, and would a technician read that as bullish or bearish?

Part D — Connecting Behavioural Finance and Technical Analysis

A junior colleague argues that technical analysis and behavioural finance are fundamentally incompatible approaches.

(i) Do you agree or disagree? Referencing LO2, explain how the disposition effect could provide a behavioural rationale for a technical momentum signal.

(ii) Identify one legitimate methodological criticism of technical analysis (from LO2) that applies regardless of whether a behavioural mechanism is proposed.

Hints & Guidance

Part A tests LO1's decision-biases table. Part B tests LO1 and Formula F1 (mirror the Royal Dutch/Shell example exactly). Part C tests LO2 and Formulas F2–F3. Part D synthesizes LO1 and LO2.